

Recommendation Systems and Media Personalization

Personalized Movie Recommendation System

Group Members
Sahil Chandrakant Suryawanshi · Saarthak Joshi · Yash Kirankumar · Nandeesh Venkatapura Siddaraju · Frederick Boafo

Summer 2026

Explore Project



Project Goal

Objective

Build a personalized movie recommender that provides better recommendations than simply suggesting popular movies to every user.

Core Methods

Baseline recommender · Content-based filtering · Collaborative filtering · Hybrid recommender



Evaluation

Top-10 recommendations · Precision & Recall metrics · HitRate & NDCG · Explainability of results



MovieLens Dataset

- **100K** ~100,000 Ratings
- **610** Users in Dataset
- **9,700** Movies Catalogued
- **3** Files: ratings.csv, movies.csv, tags.csv

Data Preprocessing

Temporal Splitting — Strategy

Training Data — 80%

Remove Data Leakage — Key

20% — Testing Data

Test — Newest Ratings

Goal — Simulates Real-World Use



Popularity-Based Recommendation

How It Works

Movies ranked by average rating and number of ratings. Bayesian weighting applied to prevent movies with very few ratings from dominating the top results.

Advantages

Simple to implement with no complex modeling required. Fast to compute and easy to interpret. Works well as a baseline for comparison with personalized methods.

Limitations

Same recommendations for every user regardless of preferences. No personalization — ignores individual taste and viewing history. Cannot adapt to niche or diverse user interests.



Content-Based Filtering

Step 1: Movie Vectorization

Each movie is transformed into a TF-IDF vector derived from its genres and user-generated tags, creating a rich semantic representation of movie content.

Step 2: User Profile

A user profile vector is computed by aggregating the TF-IDF vectors of movies the user has rated highly, capturing their individual taste and preferences.

Step 3: Cosine Similarity

Recommendations are generated by computing cosine similarity between the user profile vector and all candidate movies, returning the top-10 closest matches.

Collaborative Filtering

Item-Item Collaborative Filtering

Method

"If you liked Movie A, others who liked it also liked Movie B."

Idea

Uses rating matrix and cosine similarity to find similar movies

Similarity

Method

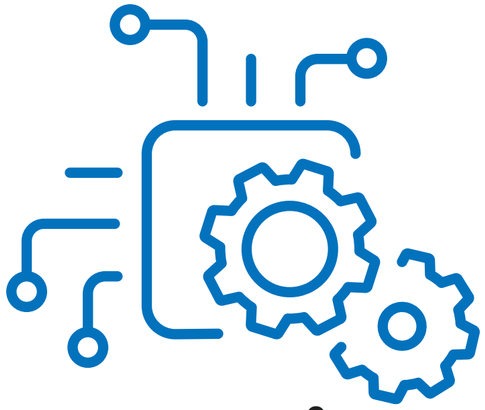
Matrix Factorization (SVD)

Idea

Discovers hidden preference factors (e.g., action lovers, comedy fans)

Advantage

Learns latent preferences — often more accurate than neighborhood methods



Hybrid Recommender

Hybrid Score

Score = α × Semantic Similarity + (1 - α) × Collaborative Preference. Combines content-based and collaborative signals into one unified recommendation score.

Key Benefits

Better accuracy and explainability. Reduces weaknesses of individual models. Balances personalization with collaborative insights.

How It Works

α controls the blend between content similarity and collaborative preference. Tuned to optimize Precision@10, Recall@10, and NDCG@10 on test data.

Evaluation Metrics



Precision@10

How many of the Top-10 recommended movies were actually relevant to the user? Measures recommendation accuracy among returned results.



Recall@10

How many of the user's relevant movies were found in the Top-10 list? Measures coverage of relevant items in recommendations.



HitRate@10 & NDCG@10

HitRate: Did we recommend at least one relevant movie?
NDCG: Were relevant movies ranked near the top of the list?

4 Top-10 Recommendations

4 Evaluation Metrics Used

80/20 Models Evaluated



Evaluation Results

- **0.38** — Hybrid Model — Best Precision@10
- **0.41** — Hybrid Model — Best NDCG@10
- **0.72** — Hybrid Model — Best HitRate@10

Top-10 Recommendation Example

🎬 Popularity List	#1-3		#1-3	🤖 Hybrid List
The Dark Knight, Forrest Gump	#4-6		#4-6	Interstellar, Ex Machina
Shawshank Redemption, Inception	#7-9		#7-9	Arrival, The Martian

Movie 1: Sci-Fi Match

Recommended because the user frequently rates Sci-Fi movies highly. Content similarity to "Interstellar" via shared genres and tags (space, science, drama). High cosine similarity in TF-IDF space.

Movie 2: Peer Preference

Recommended because highly similar users enjoyed this film. Strong SVD latent factor score aligns with user's hidden preference profile. Collaborative signal dominates.

Movie 3: Hybrid Score

Recommended by a high hybrid score combining semantic similarity and collaborative preference. Matching genres and tags reinforce both content and SVD signals.





Streamlit Demo

Select & Explore

Choose any user and select your preferred recommender model — Popularity, Content-Based, Collaborative, or Hybrid — to instantly generate personalized Top-10 movie recommendations.

View & Explain

View your Top-10 recommendations alongside detailed explanations for each suggestion — see why each movie was recommended based on genres, tags, and user similarity factors.

Compare Algorithms

Compare recommendations side-by-side across different algorithms. See how Popularity, Content-Based, and Hybrid models differ in their suggestions for the same user in real time.



System Limitations

Cold-Start Problem

New users with no rating history receive poor recommendations. Sparse ratings make it difficult to find reliable user-item similarities and degrade model performance.

No Contextual Info

The system lacks contextual signals such as time of day, user mood, or viewing occasion, which limits the relevance and situational accuracy of recommendations.

Data Quality Issues

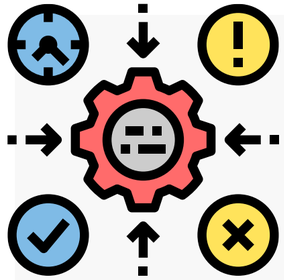
Genre categories in MovieLens are limited and coarse-grained. User-provided tags are noisy, and sparse, reducing the quality of content-based filtering.

Future Improvements



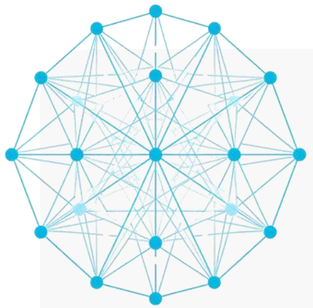
SBERT & Neural Recommenders

Replace TF-IDF with SBERT embeddings for richer semantic understanding. Incorporate deep learning and neural recommender architectures for improved accuracy and representation.



Context-Aware Recommendations

Integrate contextual signals such as time of day, user mood, and location to deliver more relevant and timely movie recommendations beyond static preferences.



GNNs & Reinforcement Learning

Use Graph Neural Networks to model complex user-movie relationships. Apply Reinforcement Learning for adaptive, continuously improving recommendations over time.

SBERT Semantic Embeddings

3+ Context-Aware Signals

GNN Graph & RL Models

Thank You

Personalized recommenders outperform popularity-based methods. Combining content-based and collaborative filtering improves recommendation quality. The Hybrid model provides the best balance of accuracy and explainability.

Questions & Discussion Welcome